Week 6 RAG Homework Starter

This starter app helps students practice Week 6 concepts from the RAG lecture:

- corpus preparation
- chunking with overlap
- simple retrieval
- answer generation with citations
- refusal behavior when evidence is missing

Learning goals

By the end of this homework, a student pair should be able to:

1. Explain how chunking strategy affects retrieval quality.
2. Implement a retrieval pipeline that returns top-k relevant chunks.
3. Return answers with explicit citations to source chunks.
4. Refuse unsupported questions instead of hallucinating.
5. Show evidence for one supported and one unsupported question.

Homework tasks for students

1. Outcome 1 (chunking impact): tune chunking parameters in `src/chunking.py` and compare retrieval quality for the same query.
2. Outcome 2 (top-k retrieval): verify `src/retrieval.py` returns correctly ordered top-k relevant chunks.
3. Outcome 3 (citations): ensure `src/rag.py` returns grounded answers with explicit citations to chunk IDs.
4. Outcome 4 (refusal): ensure unsupported questions trigger refusal instead of hallucinated answers.
5. Outcome 5 (evidence): document one supported and one unsupported query with full retrieval and output evidence.

Recommended evidence template (`evidence.md`)

Use this exact structure for easy review:

1. Outcome 1: Chunking impact
   - Config A:
   - Config B:
   - Retrieval difference summary:
2. Outcome 2: Top-k retrieval
   - Query:
   - Top-k output:
   - Why ranking is correct:
3. Outcome 3: Citation quality
   - Supported query:
   - Answer with citations:
   - Retrieved chunk IDs:
4. Outcome 4: Refusal behavior
   - Unsupported query:
   - Retrieved context:

  - Refusal output:
5. Outcome 5: Final evidence pair
  - Supported case command + output
  - Unsupported case command + output

Use `docs/homework-checklist.md` as submission guidance.